

SanthoshKumar P

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SUMMARY

Software Developer & Automation Engineer with experience in building web applications, automation solutions, and industrial software systems. Proficient in MERN Stack, Python, Django, FastAPI, and database technologies including MongoDB and MySQL. Experienced in developing robotic management platforms, UWB tracking systems, and enterprise applications. Skilled in REST API development, system integration, and AI-driven solutions. Passionate about creating scalable, secure, and efficient software for real-world business and automation challenges.

PROGRAMMING SKILLS

- **Language:** HTML5, CSS3, JavaScript, Python, Java.
- **Frameworks:** React.js, Next.js, Node.js, Express.js, Bootstrap, Tailwind CSS.
- **Database:** MongoDB, MySQL.
- **Tools :** Git, JWT, Docker, WebSockets, Postman, Power Bi, Tableau, Figma, Microsoft.
- **Design Tools:** Figma, UI/UX Prototyping.
- **AI & Automation:** YOLO, Computer Vision, AI Model Training

EXPERIENCE

Software Developer & Automation Engineer | Nov 2025 – Present

RVR Machineries Manufacturers Pvt Ltd

- Developed industrial web applications using React.js, MERN Stack, Python, Django, and FastAPI for automation, monitoring, and data management.
- Built and deployed Dustbin Management System, Fairino Robot Management System, and UWB Tracking Management System with REST API integration and database optimization.
- Developed AI-powered automation solutions using Computer Vision, YOLO, Model Training, and robotic system integration for industrial applications.
- Designed scalable backend services and APIs for real-time device communication, data processing, and operational analytics.
- Collaborated with automation and robotics teams to integrate software solutions with industrial hardware and robotic systems.

INTERNSHIPS

Incisive Designs and Business Solutions - Web Development Intern

May to June 2025

- Developed a CRM Dashboard using Next.js, Node.js, Express.js, and MySQL for efficient data management and user interaction.
- Gained hands-on experience in full-stack development and building dynamic, database-driven web applications.

Cognifyz Technologies - UI/UX Design Intern

April to May 2025

- Completed UI/UX Internship at Cognifyz Technologies & designed an Online Learning Platform using Figma.
- Focused on user-friendly layouts and interactive prototyping for a seamless learning experience.

EDUCATION

Dr. N. G. P Arts and Science College

Master of Computer Science; percentage: 88%

PSG College of Arts & Science

Bachelor of Information Technology; percentage: 86%

Coimbatore, Tamil Nadu

Jul 2024 - Apr 2026

Coimbatore, Tamil Nadu

Aug 2021 - Apr 2024

PROJECTS

Real Estate – ProjectLink: <https://github.com/21BIT035/RealEstate.git>

- Built a responsive Real Estate app using React Js, Tailwind CSS, JavaScript and MongoDB.
- Connected MongoDB for efficient property and user data management.

Whether App– ProjectLink: https://github.com/21BIT035/Whether_app.git

- Developed a responsive Weather App using React Js and Tailwind CSS with real-time data from a weather API.
- Designed a clean, user-friendly UI with seamless functionality and API key integration.

HealPay - ProjectLink: <https://github.com/21BIT035/Healpay.git>

- Developed a secure digital payment platform using React.Js, Node.Js, MongoDB and FastAPI for fast and Scalable transaction processing.
- Implemented Post-Quantum Cryptography algorithms (Kyber & Dilithium) to enhance payment security and future-proof transaction authentication.

Vehicle Warranty Management System - ProjectLink: https://github.com/21BIT035/Warranty_card.git

- Built a multi-role warranty management platform for Customers, Dealers and Admin using React.Js, Node.Js, Express.Js and MongoDB.
- Developed warranty claim processing, vehicle registration, approval workflows and role-based access control to streamline warranty operations.

Fairino Robot Management System - Python, FastAPI, React.js

- Developed an industrial robot management system to automate coffee and tea preparation using Fairino robots with configurable task execution workflows.
- Designed and implemented a React.js dashboard for order management, robot monitoring, inventory tracking, and real-time operational control.
- Integrated Razorpay payment APIs and Intel RealSense camera-based object detection for payment processing, cup identification, object localization and robot pick-and-place automation.

UWB Tracking Management System - Python, Django, React.js, PostgreSQL

- Built a real-time Ultra-Wideband (UWB) tracking system for monitoring asset and personal locations within industrial environments.
- Developed APIs, live dashboards, and location visualization features to improve tracking accuracy and operational efficiency.

Smart Dustbin Management System - Python, Django, React.js

- Developed a smart waste monitoring platform to track bin status, collection schedules and operational metrics through centralized dashboards.
- Integrated real-time monitoring, alert notifications and reporting features to improve waste management efficiency and resource utilization.

Workshops and Events

- Presented a paper on “Neural Network” at the 3rd International Conference conducted by ICT Academy.
- Secured 1st place in a Datathon competition by utilizing Power BI to create interactive data visualizations and extract meaningful insights.
- Participated in the 24-hour Hack Hive Ethical Hackathon at KPR College and the prestigious Smart India Hackathon, demonstrating strong problem-solving and innovation under pressure.

CERTIFICATIONS

- Software Testing certified by IT department of PSGCAS.
- UI/UX Designers certified by Great Learning.
- Python certified by Uniathena & Cisco Academy.
- Fundamentals of Digital marketing certified by Google Digital Garage.